PERIPHERAL & INTERFACING

Viva preparation | 300 questions and answers

MTS86 / 8086 practicals • 8255 interfaces • Wider course theory

Prepared for your uploaded Peripheral and Interfacing Lab materials (530202), Dhaka City College. The lecture slides identify the theory course as 530201.

This is a broad, syllabus-based question bank, not a promise that every possible examiner question is included. Questions are newly written study aids; they are not an official or leaked viva paper.

How to use it tonight

First master the eight arithmetic/logic programs, then 8255 configuration and the four hardware experiments. Read each answer aloud in 20-40 seconds. Cover the answer and explain the example in your own words. Use the harder follow-ups after the fundamentals feel comfortable.

Priority	Study focus	Suggested time
1	Experiments 1-8: bytes, result, binary working, flags (sections 01-04)	60 min
2	8255, 80H, port map, OUT and display/DAC/matrix (05-09)	75 min
3	Practical explanation and troubleshooting (10)	25 min
4	Interfaces, DMA, serial communication, other course devices (11-15,17)	45 min
5	Expert challenges, recap tables, spoken rehearsal (16 + appendices)	35 min

If time is very short: study sections 02-06 and 10, then the result/flag tables. Still learn the aim and basic operation of all 12 experiments because any two may be assigned. The time plan is a study suggestion, not a prediction of question frequency.

All unmarked code/data examples in the practical summaries are hexadecimal. H denotes hex, B denotes binary; 1 in SHL/SHR is the shift count.

Read these corrections first

SHR input mismatch

The contents page (PDF p.2) lists A=02H. The worked experiment (p.10) uses A=05H. Correct answers: 02H >> 1 = 01H, CF=0; 05H >> 1 = 02H, CF=1. Follow the input assigned in the exam.

Port-address mismatch

The tables on pp.11,13,15,16 use A=19H, B=1BH, C=1DH, control=1FH. The sketch on p.11 also shows 3FDx-style addressing and inconsistent labels. Do not combine the sketch with the table. Verify the actual trainer map with the demonstrated lab setup.

Display pattern assumptions

The seven-segment table in this guide uses the conventional active-low dp,g,f,e,d,c,b,a mapping. The handwritten pattern list is difficult to read. Derive patterns from the segment diagram and actual wiring; do not assume the list is universally valid.

D/A experiment ambiguity

Experiment 11 is titled D/A conversion but the written code and bar sketches do not establish the converter chip, reference voltage or analog measurement point. The general DAC answers explain the principle; the numerical 5 V example is illustrative, not a claim about your board.

Flags are not all updated alike

MOV, OUT and NOT preserve flags. AND/OR/XOR clear CF and OF but leave AF undefined. SHL/SHR have count-dependent overflow rules. Do not report an undefined or preserved flag as newly calculated.

PIC has two contexts

Lecture 04 discusses Microchip PIC microcontrollers. In 8086 interrupt-controller questions, PIC can mean Programmable Interrupt Controller (8259). State which context you mean.

Minor theory slips

CMYK means cyan, magenta, yellow and black. LEA is the 8086 Load Effective Address mnemonic, not LDA. LCDs do refresh/update. A CD pit is not directly one user-data bit; optical transitions undergo decoding.

Question-bank map

01. 8086 and trainer-board foundations | Q1-26 | Basic
02. Experiments 1 and 2: ADD and SUB | Q27-42 | Basic to intermediate
03. Experiments 3-6: AND, OR, NOT and XOR | Q43-62 | Basic to intermediate
04. Experiments 7 and 8: shifts and flags | Q63-82 | Intermediate to expert
05. 8255 PPI and port programming | Q83-111 | Basic to expert
06. Experiment 9: seven-segment display | Q112-127 | Basic to expert
07. Experiment 10: LED display | Q128-138 | Basic to expert
08. Experiment 11: digital-to-analog conversion | Q139-154 | Basic to expert
09. Experiment 12: LED matrix | Q155-166 | Basic to expert
10. Program tracing and practical troubleshooting | Q167-185 | Intermediate to expert
11. Interface units, buses, interrupts and DMA | Q186-211 | Basic to expert
12. Serial and parallel communication | Q212-227 | Basic to expert
13. Keyboard, encoders, stepper motor and relay | Q228-242 | Basic to expert
14. Data entry and display devices | Q243-253 | Basic to intermediate
15. Printers, plotters and storage | Q254-270 | Basic to intermediate
16. Fast oral challenges | Q271-280 | Expert
17. Additional questions from the class notes | Q281-300 | Intermediate to expert

Level guide: Basic = definitions and direct results. Intermediate = explanation, decoding and application. Expert = edge cases, flags, timing, troubleshooting and design tradeoffs. A section may progress through more than one level.

Source map

Primary scope: 1-2. PI-R(1).pdf, PDF pp.2-18, including all 12 practicals. Lecture scope: PI 01-11 inside slides-pi(1).zip. Class-note scope: the seven chapter PDFs and two question-sheet images inside class-notes-pi(1).zip. Page references in this guide use PDF page positions, not handwritten numbering.

The question bank combines those topics with standard explanations and calculated examples. Some scanned notes are hard to read; no exact board key sequence, unclear DAC connection or uncertain segment mapping is invented.

01 / 8086 and trainer-board foundations

Basic

1. What is this practical course about?

It is Peripheral and Interfacing Lab (530202). We use an MTS86 trainer to execute 8086 machine-code programs and control external displays and related circuits.

2. What is a peripheral?

A device providing input, output, storage or another supporting function, such as a keyboard, display or printer.

3. What is interfacing?

Connecting devices so they can exchange information correctly despite differences in timing, voltage, format and operating speed.

4. Why is an interface needed between CPU and a peripheral?

It provides address selection, data buffering, status/control signals and suitable timing or electrical conversion.

5. What is a microprocessor?

A CPU implemented on an integrated circuit. It fetches, decodes and executes instructions and usually needs external memory and I/O circuitry.

6. How is a microcontroller different?

A microcontroller integrates a CPU, memory and peripherals on one chip; a microprocessor-based system commonly uses more external components.

7. Why is the 8086 called a 16-bit processor?

Its main registers, ALU operations and external data bus support 16-bit data. Its address bus is 20 bits, not 16 bits.

8. How much memory can the 8086 address?

2^20 bytes = 1 MiB, from physical address 00000H to FFFFFH.

9. What are the main units of the 8086?

The Execution Unit executes instructions. The Bus Interface Unit handles bus transfers, address formation and instruction prefetching.

10. What is the instruction queue?

The 8086 BIU can prefetch up to six instruction bytes, allowing fetching and execution to overlap. A taken branch redirects fetching and discards obsolete queued bytes.

11. What are AX, BX, CX and DX?

General-purpose 16-bit registers, conventionally used as accumulator, base, count and data registers. Some instructions have specific implicit uses for them.

12. What is AL?

The low 8 bits of AX; AH is the high 8 bits. AX = AH x 256 + AL.

13. If AX=1234H and MOV AL,56H executes, what is AX?

1256H. Writing AL does not clear or overwrite AH.

14. What are CS, DS, SS and ES?

Code, data, stack and extra segment registers. They supply segment bases for memory addressing.

15. What is IP?

The Instruction Pointer is the offset of the next instruction within the current code segment; instruction fetching uses CS:IP.

16. What are SP, BP, SI and DI?

Stack Pointer, Base Pointer, Source Index and Destination Index. They support stack, addressing and string operations.

17. What is physical address formation?

Physical address = segment x 10H + offset, using the 8086's 20 address lines. For example, 2000H:1234H gives 21234H.

18. Why use hexadecimal?

One hex digit represents four bits, making binary instruction bytes and addresses compact to enter and inspect.

19. What are bit, nibble, byte and word?

A bit is one binary digit; a nibble is 4 bits; a byte is 8 bits; an 8086 word is 16 bits.

20. What is machine language versus assembly language?

Machine language is encoded bytes executed by the CPU. Assembly uses readable mnemonics such as MOV and ADD; an assembler converts them to machine code.

21. What are opcode and operand?

An opcode specifies the operation; operands specify data or locations. In B0 02, B0 encodes MOV AL,immediate and 02 is the immediate value.

22. What is immediate addressing?

The operand value is encoded in the instruction itself, as in MOV AL,02H.

23. How do immediate, register and memory operands differ?

MOV AL,02H loads a constant; MOV AL,BL copies a register; MOV AL,[1234H] reads a byte from memory using the default data segment.

24. What does a trainer monitor do?

Its firmware helps enter and inspect memory, inspect registers and run or single-step programs. Exact keys and commands depend on the trainer version.

25. What is the difference between address and data?

An address selects a location or port; data is the value stored or transferred. In OUT 19H,AL, 19H is the port address and AL supplies data.

26. What does single-stepping help you verify?

One instruction at a time, I can inspect the next IP, destination register and affected flags to locate an error.

02 / Experiments 1 and 2: ADD and SUB

Basic to intermediate

27. Explain the addition experiment.

Load AL with 01H, then add F0H. The result is F1H in AL: 1 + 240 = 241 in unsigned decimal.

28. What are its assembly and machine bytes?

MOV AL,01H = B0 01; ADD AL,0F0H = 04 F0. Together: B0 01 04 F0.

29. Why write 0F0H in assembly?

Many Intel-style assemblers require a leading digit for hex constants beginning with A-F. A trainer's hex-entry interface simply accepts F0 as a byte.

30. Where is the addition result stored?

In AL, because ADD writes to its first operand. It is not automatically stored in memory or sent to a display.

31. What flags result from 01H+F0H?

CF=0, OF=0, SF=1, ZF=0, PF=0 and AF=0. F1H has five set bits, so even-parity flag PF is clear.

32. Why is SF=1 without signed overflow?

SF copies the result's most significant bit. Signed F0H is -16, so 1+(-16)=-15, which fits in an 8-bit signed value; OF=0.

33. Explain the subtraction experiment.

MOV AL,07H followed by SUB AL,02H gives AL=05H. Machine bytes: B0 07 2C 02.

34. What flags result from 07H-02H?

CF=0, OF=0, SF=0, ZF=0, PF=1 and AF=0. 05H has two set bits.

35. What happens for 02H-07H?

AL becomes FBH, the 8-bit two's-complement representation of -5. CF=1 indicates unsigned borrow; OF=0 because signed -5 is representable.

36. What happens for FFH+01H?

AL becomes 00H and CF=1. ZF=1, OF=0, SF=0, PF=1, AF=1. The mathematical unsigned result is 100H, but AL stores only eight bits.

37. What happens for 7FH+01H?

AL becomes 80H, CF=0 and OF=1. Signed 127+1 cannot fit in the range -128 to +127.

38. How do ADD and ADC differ?

ADD computes destination+source. ADC also includes the current carry flag, allowing arithmetic across multiple bytes or words.

39. How do SUB and SBB differ?

SUB subtracts the source. SBB subtracts both the source and the current carry flag, propagating a borrow through multi-byte arithmetic.

40. How do CMP and SUB differ?

Both calculate subtraction flags, but CMP discards the result and leaves its operands unchanged.

41. Do MOV and OUT change arithmetic flags?

No. Existing arithmetic flags are preserved, so flags seen after MOV alone may come from an earlier operation.

42. How do INC and ADD value,1 differ?

Both increment the value, but INC preserves CF. ADD updates CF along with the other arithmetic flags.

03 / Experiments 3-6: AND, OR, NOT and XOR

Basic to intermediate

43. What is the AND rule?

An output bit is 1 only if both corresponding input bits are 1. Truth table for 00,01,10,11: 0,0,0,1.

44. Explain the AND experiment.

02H=00000010B and A3H=10100011B. Their AND is 00000010B=02H. Code: B0 02 24 A3.

45. What is the OR rule?

An output bit is 1 if either or both input bits are 1. Truth table for 00,01,10,11: 0,1,1,1.

46. Explain the OR experiment.

02H OR A3H gives A3H because A3H already has bit 1 set. Code: B0 02 0C A3.

47. Is Boolean A+B the same as arithmetic addition?

No. In Boolean algebra '+' may mean OR. Here 02H OR A3H=A3H, whereas 02H+A3H=A5H.

48. What is NOT?

Bitwise inversion: each 0 becomes 1 and each 1 becomes 0, across the operand's width.

49. Explain the NOT experiment.

NOT of 8-bit 02H gives FDH: 00000010B becomes 11111101B. Code: B0 02 F6 D0.

50. Why is the result FDH rather than FFFDH?

AL is eight bits. NOT AX on AX=0002H would produce the 16-bit result FFFDH.

51. Does NOT affect the flags?

No. On the 8086, NOT leaves flags unchanged even though it changes the operand.

52. What is the XOR rule?

An output bit is 1 when the input bits differ. Truth table for 00,01,10,11: 0,1,1,0.

53. Explain the XOR experiment.

02H XOR A3H gives A1H: 00000010B XOR 10100011B=10100001B. Code: B0 02 34 A3.

54. Which flags do AND, OR and XOR affect?

They clear CF and OF; set SF, ZF and PF from the result; leave AF undefined. Do not give a definite AF value.

55. What are the key flags for the three lab results?

AND result 02H: SF=0,ZF=0,PF=0. OR result A3H: SF=1,ZF=0,PF=1. XOR result A1H: SF=1,ZF=0,PF=0. CF=OF=0 in each.

56. How can you clear AL quickly?

XOR AL,AL gives zero, with ZF=1 and CF=OF=0. MOV AL,00H also clears AL but preserves flags.

57. How do you clear a particular bit?

AND with a mask having 0 at that bit and 1 elsewhere. AND AL,0FEH clears bit 0.

58. How do you set a particular bit?

OR with a mask having 1 at that bit. OR AL,08H sets bit 3 without changing the others.

59. How do you toggle a particular bit?

XOR with a mask having 1 at that bit. XOR AL,08H toggles bit 3.

60. What is TEST and how is it useful?

TEST performs an AND for flags without storing the result. TEST AL,08H followed by a ZF check tests whether bit 3 is set.

61. Is NOT the same as NEG?

No. NOT produces one's complement. NEG computes 0 minus the operand, equivalent to two's complement. For 02H, NOT gives FDH; NEG gives FEH.

62. What identities should you remember?

A AND 00=00; A AND FF=A; A OR 00=A; A OR FF=FF; A XOR 00=A; A XOR A=00, for an 8-bit A.

04 / Experiments 7 and 8: shifts and flags

Intermediate to expert

63. What does SHL do?

It shifts bits left, fills the least significant bit with 0 and copies the bit shifted out into CF.

64. Explain the SHL experiment.

MOV AL,02H; SHL AL,1 gives 04H. Bytes: B0 02 D0 E0. The second operand is a shift count, not another number to add.

65. What does SHR do?

It shifts bits right, fills the most significant bit with 0 and copies the last bit shifted out into CF.

66. What is the SHR result for the contents-page input?

The contents page specifies 02H. 02H SHR 1 gives 01H, CF=0. Bytes: B0 02 D0 E8.

67. What is the SHR result for the worked-page input?

PDF page 10 uses 05H. 05H SHR 1 gives 02H, CF=1 because the original bit 0 was 1. Bytes: B0 05 D0 E8.

68. Does SHL always multiply by 2 correctly?

It multiplies the bit pattern modulo 2^n. It matches the full mathematical result only when no significant information is lost and signed interpretation is handled correctly.

69. Does SHR divide by 2?

For an unsigned operand, SHR by one gives the integer quotient rounded down; the discarded low bit goes into CF.

70. What happens when 80H is shifted left once?

Result=00H, CF=1 and OF=1. The original top bit is lost and the signed result is not representable.

71. What happens when FFH is shifted right once?

SHR gives 7FH and CF=1. It treats the bit pattern as unsigned for division.

72. How does SAR differ from SHR?

SAR repeats the sign bit instead of inserting 0. For FFH, SAR by one gives FFH, preserving signed -1; SHR gives 7FH.

73. How does a rotate differ from a shift?

ROL/ROR move the shifted-out bit back into the opposite end. RCL/RCR rotate through CF. Logical shifts insert zeros instead.

74. What are the count options on the original 8086?

A shift or rotate uses count 1 or the value in CL. For multiple shifts use, for example, MOV CL,3 followed by SHL AL,CL; do not assume later immediate-count encodings exist.

75. How is OF defined for a one-bit SHL?

OF is the XOR of the new most significant bit and CF. For counts greater than one, OF is undefined.

76. How is OF defined for a one-bit SHR?

OF equals the original most significant bit. For counts greater than one, OF is undefined.

77. Which other flags do shifts affect?

For nonzero counts, SF, ZF and PF reflect the result, CF holds the last shifted-out bit and AF is undefined. For count zero, flags and operand are unchanged.

78. What is the difference between CF and OF?

CF reports unsigned carry or borrow; OF reports a signed result outside the operand's representable range. They answer different questions.

79. What is PF?

PF=1 when the low byte of the result has an even number of set bits, including zero set bits. It does not mean the numeric value is even.

80. What is AF?

Auxiliary carry indicates a carry/borrow across the bit-3 to bit-4 boundary in relevant arithmetic operations; it supports decimal-adjust operations.

81. What are TF, IF and DF?

TF enables single-step traps; IF controls recognition of maskable hardware interrupts; DF controls increment/decrement direction for string instructions.

82. What is the eight-bit signed and unsigned range?

Unsigned: 0 to 255. Two's-complement signed: -128 to +127. The same bits can have either interpretation.

05 / 8255 PPI and port programming

Basic to expert

83. What is PPI?

Programmable Peripheral Interface. The 8255 provides programmable parallel I/O between a processor and external devices.

84. How many I/O lines does the 8255 have?

24, arranged as three 8-bit ports: A, B and C. Port C can also be treated as upper and lower four-bit groups.

85. What are Group A and Group B?

Group A controls Port A and upper Port C; Group B controls Port B and lower Port C, subject to the chosen handshake mode.

86. What is Mode 0?

Simple input/output without handshaking. Outputs are latched; inputs are not latched. It suits straightforward LEDs and display outputs.

87. What is Mode 1?

Strobed input/output. Port A and/or B use selected Port C bits for handshake and status signals.

88. What is Mode 2?

Bidirectional strobed bus operation for Port A, with Port C handshake signals. Port B does not support Mode 2.

89. What is BSR mode?

Bit Set/Reset mode changes one selected Port C bit through the control register, without rewriting an entire Port C data byte.

90. Why do these display programs begin with 80H?

80H=10000000B selects the I/O control format, Mode 0 for both groups and outputs for Port A, Port B and both halves of Port C.

91. What does each I/O-mode control bit mean?

D7=1 selects I/O mode; D6-D5 select Group A mode; D4 sets A direction; D3 upper C direction; D2 Group B mode; D1 B direction; D0 lower C direction. Direction 1=input, 0=output.

92. Decode 83H.

10000011B means Mode 0, A output, upper C output, B input and lower C input.

93. Decode 92H.

10010010B means Mode 0, A input, B input and both halves of C output.

94. What control word gives all inputs in Mode 0?

9BH: D4,D3,D1,D0 are all 1; mode bits remain zero; D7 is 1.

95. What happens to the ports on reset?

They are configured as inputs. The program must configure the required output modes before trying to drive LEDs or other loads.

96. How is a BSR control word formed?

D7=0; D3-D1 select Port C bit 0-7; D0 selects set=1 or reset=0. D6-D4 are don't-care bits, usually written as zero.

97. What words set and reset PC5?

Set PC5: 0BH. Reset PC5: 0AH. In general, choose bit number k and write 2k+1 to set or 2k to reset.

98. What words set and reset PC7?

Set PC7: 0FH. Reset PC7: 0EH.

99. What is the port map in the handwritten program tables?

Port A=19H, Port B=1BH, Port C=1DH, control=1FH. These are board-mapped addresses, not universal 8255 addresses.

100. What does OUT 1FH,AL do?

It writes AL to the I/O port mapped at 1FH; in these tables that is the 8255 control register. With AL=80H it configures Mode 0 outputs.

101. Why are the listed ports two addresses apart?

This is consistent with an 8086 byte-lane mapping where CPU address lines above A0 drive the 8255 register selects. Confirm the actual board wiring; address spacing is not intrinsic to every 8255 system.

102. How does the 8255 select its internal registers?

With the chip selected, its own A1:A0 values 00,01,10,11 select A,B,C and control respectively. These are chip pins, not necessarily identically numbered CPU address lines.

103. What do CS, RD, WR and RESET do?

Active-low CS selects the chip; active-low RD reads data; active-low WR writes data/control; active-high RESET returns ports to input configuration.

104. What is a latch and why is it useful here?

A latch retains an output value after the CPU write ends, so a static display need not be continuously rewritten.

105. Can I use MOV to access these I/O ports?

In isolated I/O, use IN/OUT. MOV accesses memory; it can access an interface only when that interface is memory-mapped.

106. How do immediate-port and DX-port OUT differ?

OUT imm8,AL addresses ports 00H-FFH. OUT DX,AL uses the full 16-bit address in DX and can address any byte port in the 64 KiB I/O space.

107. What machine bytes encode these OUT forms?

E6 followed by an 8-bit port number encodes OUT imm8,AL; EE encodes OUT DX,AL.

108. How would you output 1FH to port 3FD8H?

MOV DX,3FD8H; MOV AL,1FH; OUT DX,AL. Bytes: BA D8 3F B0 1F EE. Use this only if the board's actual map assigns that port.

109. Why must you not mix the two maps in the lab scan?

The page-11 sketch shows a 3FDx-style address/code while its table uses 19H-1FH. These are not interchangeable; use the instructor-approved map for the actual trainer.

110. Can every 8255 port bit independently select input/output?

No. Mode 0 direction is selected by whole Port A, whole Port B and each half of Port C. BSR is individual bit set/reset, not arbitrary per-bit direction selection.

111. Can the 8255 directly drive any load?

No. Respect its current and voltage limits. Displays, relays and motors may require resistors, buffers or transistor/driver stages.

06 / Experiment 9: seven-segment display

Basic to expert

112. What is a seven-segment display?

Seven independently controlled LED segments labelled a-g form numerals and some letters. A separate decimal point may make eight controllable LEDs.

113. What is common anode versus common cathode?

Common anode ties LED anodes together; segments typically turn on when their cathodes are driven low. Common cathode ties cathodes together; segments typically turn on when their anodes are driven high.

114. Why does C0H commonly display zero?

With active-low segments and bit order dp,g,f,e,d,c,b,a, C0H=11000000B lights a-f and turns off g and dp. This depends on actual wiring and drivers.

115. Which segments display 1, 7 and 8?

1 uses b,c; 7 uses a,b,c; 8 uses all a-g. Decimal point remains independently controlled.

116. What is the lab program for displaying zero?

From the table: write 80H to control 1FH; F0H to B at 1BH; 00H to C at 1DH; C0H to A at 19H. Confirm jumpers and board map before execution.

117. What are its exact bytes?

B0 80 E6 1F B0 F0 E6 1B B0 00 E6 1D B0 C0 E6 19. These implement the table, not the conflicting sketch beneath it.

118. Which byte changes when I want another character?

Change the immediate segment-pattern byte in MOV AL,pattern before OUT 19H,AL. Do not accidentally change the B0 opcode or the 19H port address.

119. Why not write numeric 05H to display the digit 5?

A directly driven display needs a segment pattern, not the numeric value. With the stated active-low mapping, digit 5 uses 92H. A decoder would perform the conversion for you.

120. What are the usual active-low hexadecimal patterns?

Under dp,g,f,e,d,c,b,a mapping:
0=C0,1=F9,2=A4,3=B0,4=99,5=92,6=82,7=F8,8=80,9=90,A=88,b=83,C=C6,d=A1,E=86,F=8E.
Glyph variants and wiring can differ.

121. How do common-cathode patterns relate?

With identical bit order and direct noninverting drivers, complement each active-low byte. Zero changes from C0H to 3FH. Verify the decimal-point bit too.

122. Does the shown program automatically count from 0 to F?

No. It writes one selected pattern. Automatic counting requires a lookup table or repeated pattern writes, delays and loop control.

123. Why use a lookup table?

It maps a number to its segment pattern, avoiding repeated bit calculations and making 0-F sequences easy to implement.

124. How does XLAT help?

On the 8086, XLAT replaces AL with the byte at DS:[BX+AL]. Put the table base in BX and the character index in AL, with DS correctly initialized.

125. What is multiplexing?

Repeatedly enable one digit, output its pattern, then move to the next digit fast enough to appear continuous. It saves segment wiring and drivers.

126. Why might a display flicker or show ghosting?

Slow scanning causes flicker; changing segment data while the wrong digit is enabled can cause ghosting. Blank digits during transitions and scan at a suitable rate.

127. What is the 7447 used for?

It is a BCD-to-seven-segment decoder/driver normally used with common-anode displays and active-low outputs. Its defined decimal function is for BCD 0-9; it is not a general hexadecimal A-F decoder.

07 / Experiment 10: LED display

Basic to expert

128. What is the aim of the LED experiment?

To send digital bit patterns through the interface and observe the corresponding LED states, showing software-controlled output.

129. What sequence appears in the experiment table?

Configure 80H at 1FH; write FFH to Port A, 00H to Port C and F0H to Port B. The manual also shows a jumper setting; use the demonstrated lab configuration.

130. What are the corresponding bytes?

B0 80 E6 1F B0 FF E6 19 B0 00 E6 1D B0 F0 E6 1B.

131. Does a logic 1 always mean an LED is on?

No. Active-high wiring lights at 1; active-low wiring lights at 0. Driver inversion can change the relationship further.

132. Why should you not memorize the drawn LED order blindly?

Bit-to-LED position, colour and polarity depend on the board wiring and jumper selection. Test a single-bit pattern against the actual hardware.

133. How can you test LED mapping?

With the correct polarity, move a single active bit across the byte and record which LED responds. Use a one-zero pattern for active-low wiring.

134. What are useful test patterns?

00H and FFH test opposite extremes; AAH=10101010B and 55H=01010101B test alternating bits. Their visible meaning depends on polarity.

135. How would you make a running LED?

Output a one-hot pattern, delay, shift or rotate it, and repeat. For active-low LEDs, output the complemented pattern.

136. Why is a delay required?

The processor changes patterns much faster than human vision. A delay makes each visible state last long enough to observe.

137. Why are current-limiting resistors used?

An LED has a nonlinear current-voltage characteristic. A resistor or controlled-current driver limits current and protects the LED and output circuitry.

138. How do you estimate an LED resistor?

R approximately equals (supply voltage - LED forward voltage - driver voltage drop) / desired current. Check the ratings and available current before selecting it.

08 / Experiment 11: digital-to-analog conversion

Basic to expert

139. What is a DAC?

A Digital-to-Analog Converter produces an analog voltage or current corresponding to a digital input code.

140. How does an ADC differ?

An ADC converts an analog input into a digital code. A DAC performs the reverse direction.

141. What does your experiment-11 listing actually show?

It configures the 8255 and writes FFH to A, F0H to B and 00H to C; the instruction beneath varies the final value. It does not clearly identify the DAC chip, reference or analog measurement connection.

142. Can the drawn bar display alone prove D/A conversion?

No. It shows a visual response but does not establish an analog transfer function. Verify the trainer circuit and measure the actual analog output if that is part of the assigned experiment.

143. What can you safely explain about varying its input?

Changing the data code changes the intended output level through the connected hardware. The exact voltage, polarity and scale require the actual DAC and reference details.

144. How many codes does an 8-bit DAC have?

256, from 0 to 255. There are 255 transitions between its lowest and highest codes.

145. What is resolution?

The smallest ideal output step for a one-code change. For an ideal unipolar voltage DAC with scale Vref, step size is Vref/2^n.

146. What is the ideal unipolar DAC equation?

In a common convention, Vout=Vref x D/2^n, where D is the unsigned code. At the maximum code, Vout is one LSB below Vref. Actual circuits can have gain or inversion.

147. What values result for an ideal 8-bit DAC with Vref=5 V?

Step=5/256=19.53125 mV; D=0 gives 0 V; D=128 gives 2.5 V; D=255 gives approximately 4.98047 V.

148. When would a formula use 255 instead of 256?

If VFS is defined as the actual output at the highest code, use Vout=VFS x D/255 for an ideal 8-bit endpoint scale. State the definition; Vref and endpoint full-scale output are not automatically the same.

149. What if the trainer uses a DAC0808?

It is an 8-bit current-output DAC: ideally Iout=Iref x D/256. A suitable external circuit converts current to voltage. The scan does not establish that this is your exact component.

150. What is settling time?

The time after a code change for the analog output to enter and stay within a specified error band around its final value.

151. What are resolution and accuracy?

Resolution is step size or number of distinct codes. Accuracy is closeness to the ideal output; more bits alone do not guarantee better accuracy.

152. What are offset, gain and linearity errors?

Offset shifts the transfer curve; gain error changes its slope; nonlinearity makes points deviate from the ideal straight-line relationship after the relevant offset/gain convention.

153. How do you generate a ramp, square wave or sine wave?

Ramp: repeatedly increase codes. Square wave: alternate two codes. Sine wave: output sampled sine values from a table. Timing and reconstruction filtering affect the waveform.

154. What is the output frequency for a repeating N-sample table?

Ideally f_wave=f_update/N. Include program and I/O overhead when determining the actual update rate.

09 / Experiment 12: LED matrix

Basic to expert

155. What is an LED matrix?

LEDs arranged in rows and columns, allowing characters and graphics to be created by selecting intersecting lines.

156. How many LEDs are in an 8x8 matrix?

64 per colour plane. A two-colour matrix may contain two LED elements at each position.

157. What is the aim of your matrix experiment?

Configure output ports and select a row/column pattern to light a line on the trainer matrix; the notes discuss red and green through different port settings.

158. What sequence is shown?

Configure 80H at 1FH; write FFH to A at 19H; 00H to B at 1BH; 00H to C at 1DH. A later worked sketch changes Port C to 01H for a selected line.

159. Can we derive every row and colour from the code alone?

No. We need the decoder, polarity and wiring. Explain the port values separately from the observed board-specific row and colour.

160. How do you display a full character?

Store one pattern per row or column and repeatedly scan them with matching select signals. The scan reconstructs the complete glyph visually.

161. Why may the static lab program light only one line?

It writes only one set of select/data values. Displaying a complete arbitrary character normally requires several rows/columns to be refreshed in sequence.

162. What is duty cycle in an eight-row scan?

With one row enabled for equal time and no blanking overhead, each row has duty cycle 1/8=12.5%.

163. How are scan rate and frame rate related?

If each of eight rows is selected 800 times per second in sequence, one full frame repeats 100 times per second: frame rate=row-update rate/8.

164. What causes ghosting and how do you reduce it?

Old data can appear on a newly selected row, or vice versa. Disable output, update data and selection, then enable the intended row.

165. How would you move a pattern horizontally?

Shift the stored row bit patterns left or right, handle edge bits as intended, and keep scanning the rows.

166. Why does brightness change during multiplexing?

Brightness depends on peak current, duty cycle and device characteristics. Do not compensate by exceeding LED or driver peak-current ratings.

10 / Program tracing and practical troubleshooting

Intermediate to expert

167. What should you say when asked to explain any experiment?

State the aim, input, relevant hardware, each instruction's action, expected result, affected flags or output pattern, and how you verified it.

168. How do you enter and test a short arithmetic program?

Select the instructor-approved RAM address, enter bytes consecutively, read them back, set the correct execution address, single-step the intended instructions and inspect AL and flags.

169. If a program starts at offset 1000H, where does B0 01 04 F0 go?

Put B0 at 1000H, 01 at 1001H, 04 at 1002H and F0 at 1003H. Instruction boundaries are 1000H and 1002H.

170. What if execution starts at the immediate-data byte?

The CPU treats that byte as an opcode, so execution differs from the intended program. Always start at an instruction boundary.

171. Why can a correct two-instruction program still go wrong in run mode?

After the last instruction, the CPU continues into following memory unless stopped or redirected. Use the monitor's approved stop, breakpoint or return method.

172. Is HLT the same as program exit?

No. HLT pauses instruction execution until an appropriate interrupt or reset condition resumes it; it is not a universal return to the trainer monitor.

173. Why should you not append RET blindly?

RET expects a valid return address on the stack. It is correct only if the monitor invoked the program through a compatible call convention.

174. What if the expected result is in AL but the display is unchanged?

Arithmetic does not automatically output to a peripheral. Inspect AL through the monitor or explicitly convert and send the value to the display interface.

175. Why is B0 02 not reversed by little-endian storage?

B0 is the opcode and 02 is a one-byte operand. Little endian applies to multi-byte numeric fields, not reversal of the entire instruction stream.

176. What does BA D8 3F mean?

MOV DX,3FD8H. The two-byte immediate is stored low byte first, D8 then 3F.

177. What if the LED/display remains blank?

Check power and connections, approved jumper selection, execution address, entered bytes, control word, port map, output-enable/selection signals and active-high/low polarity.

178. What if the wrong segments light?

Check the segment bit order, polarity, lookup-table byte, decoder assumptions and whether the data reached the intended port.

179. What if all ports are programmed as inputs?

OUT writes will not drive the external pins as required for the display. Recheck the control word and whether it was written to the control port.

180. What if the display is correct briefly and then changes?

The CPU may be running into unintended bytes, a scan routine may overwrite the port, or an interrupt/monitor routine may reuse the interface.

181. How do you prove the code works for more than one input?

Test an ordinary case and boundary cases: zero, FFH, carry/borrow, sign boundary, or one-bit output patterns, depending on the experiment.

182. What if sir changes A or B?

Recompute in binary/hex, replace only the immediate operand bytes, and predict the result and flags before execution.

183. Why might a software delay vary?

CPU clock, wait states, instruction timing and interrupts affect it. A hardware timer is usually better for stable timing.

184. How can you preserve unrelated port bits?

Maintain a software copy of the output byte, modify only selected bits with masks and write the result. Avoid assuming every port read returns its output latch.

185. Why use a software shadow instead of repeatedly reinitializing the 8255?

A shadow preserves intended output values; mode-setting writes can reset output latches and disrupt active outputs. Configure once, then update data or use suitable BSR operations.

11 / Interface units, buses, interrupts and DMA

Basic to expert

186. What are the main parts of an interface unit?

Data register, status register, control register, address decoder/comparator and control logic.

187. What do data, status and control registers do?

Data registers hold transferred values; status registers report conditions such as ready/busy; control registers accept commands or configuration.

188. What are the address, data and control buses?

Address lines select locations; data lines carry values; control lines indicate operations and timing, such as read/write and interrupt acknowledgement.

189. What does 'highway' mean in your notes?

It is a bus or shared communication path linking the CPU, memory and interface units.

190. What is the benefit and drawback of a shared bus?

It reduces connections and cost, but devices compete for bandwidth and require selection/arbitration to avoid conflicts.

191. What is bus contention?

Two devices drive incompatible values onto the same bus simultaneously. Correct chip selection and tri-state control prevent it.

192. What is tri-state output?

An output can drive 0, drive 1 or enter high impedance, allowing another device to drive a shared line.

193. What is address decoding?

Hardware recognizes a device's assigned address and generates its chip-select signal.

194. What is partial versus full address decoding?

Full decoding checks all relevant address bits; partial decoding checks fewer and can make one device respond at multiple alias addresses.

195. What is programmed I/O?

CPU instructions explicitly move data between the processor and a peripheral. In polling-based programmed I/O, the CPU repeatedly checks readiness.

196. What is polling?

Repeatedly reading status until a condition is met. It is simple but can waste CPU time while waiting.

197. What is interrupt-driven I/O?

The device requests service when needed; the CPU temporarily executes an interrupt service routine, reducing repeated status checks.

198. What is an ISR?

An Interrupt Service Routine handles an interrupt, services/acknowledges its source and restores the interrupted program's context as required.

199. What is a vectored interrupt?

Its type or vector identifies the service routine's entry address. The 8086 uses an interrupt vector table with 256 entries of four bytes each.

200. Where is the 8086 interrupt vector table?

At physical 00000H-003FFH. For interrupt type n, its pointer starts at address 4n and contains IP followed by CS.

201. How do INTR and NMI differ on the 8086?

INTR is maskable through IF and obtains its interrupt type through acknowledgement. NMI is non-maskable by IF and uses type 2.

202. What is IRET?

Interrupt return restores IP, CS and FLAGS from the stack. It is different from an ordinary near RET.

203. What is interrupt priority or daisy chaining?

Priority selects which request is serviced first. A daisy chain passes acknowledgement through devices in order, often giving earlier devices higher priority.

204. What is a nested interrupt?

A new interrupt is serviced before the earlier ISR finishes, when masking/priority and software permit it. Context and stack use must be managed carefully.

205. What is DMA?

Direct Memory Access lets a controller transfer data between a device and memory without the CPU moving each item. The CPU still normally sets up and supervises the transfer.

206. What does a DMA controller need?

Typically a starting address, transfer count, direction/mode and a device request. It arbitrates for the bus and updates address/count as transfers proceed.

207. What is cycle stealing versus burst DMA?

Cycle stealing uses individual bus opportunities and interleaves CPU access; burst DMA keeps the bus for a block, increasing transfer efficiency but delaying CPU bus use.

208. Why might DMA get priority?

A peripheral may have real-time data deadlines and limited buffering. Giving it timely bus access prevents overrun/underrun; priority is a system policy, not a universal guarantee.

209. How do memory-mapped and isolated I/O differ?

Memory-mapped devices occupy memory addresses and use memory instructions. Isolated I/O occupies a separate port space and uses IN/OUT on the 8086.

210. What is PIC and what is DMAC?

PIC is context-dependent: in an 8086 interrupt system it means Programmable Interrupt Controller, such as 8259. Lecture 04 instead discusses Microchip PIC microcontrollers. Identify the context before expanding the term. DMAC means Direct Memory Access Controller.

211. What is PCI?

Peripheral Component Interconnect, a peripheral bus standard. Conventional PCI is parallel; PCI Express is a different serial, packet-based interconnect.

12 / Serial and parallel communication

Basic to expert

212. What is serial versus parallel transmission?

Serial sends bits sequentially per lane; parallel sends several bits on separate lines simultaneously.

213. Is parallel always faster?

No. Clock rate, lane count, signalling, skew, distance and protocol overhead determine throughput. High-speed serial links can outperform parallel buses.

214. What are simplex, half duplex and full duplex?

Simplex is one-way; half duplex allows both directions but not simultaneously; full duplex allows simultaneous two-way transfer.

215. What is asynchronous serial communication?

Characters are framed with start/stop bits and optional parity, without a separate shared continuous clock line. Sender and receiver must agree on timing.

216. What is synchronous communication?

The receiver has or recovers timing synchronized with the transmitted bit stream; data can be grouped into frames without a start/stop pair for every character.

217. What is 8N1?

Eight data bits, no parity, one stop bit; with the start bit, each character occupies ten bit periods.

218. What payload rate does 9600 bit/s give with 8N1?

At most 960 characters/second before additional protocol overhead: 9600/10. Eight-bit payload throughput is 7680 bit/s.

219. What is baud versus bit rate?

Baud is symbols per second; bit rate is bits per second. They are equal only when each symbol carries one bit.

220. What is parity and what can it miss?

A parity bit makes the total number of ones even or odd. It detects an odd number of bit errors but can miss even-numbered error changes and does not correct them.

221. What are framing and overrun errors?

Framing error means the expected stop-bit condition was not seen. Overrun means new data arrived before earlier received data was handled or buffered adequately.

222. What is handshaking?

Control signals coordinate readiness and completion so devices of different speeds can transfer data reliably.

223. What is UART versus USART?

A UART handles asynchronous serial communication. A USART supports synchronous as well as asynchronous operation; the 8251A is a USART.

224. What is RS-232?

A serial electrical/interface standard traditionally connecting DTE and DCE equipment. It is not directly compatible with ordinary TTL voltage levels; a suitable transceiver is required.

225. What are DTE, DCE and null modem?

DTE is terminal/computer equipment; DCE is communication equipment such as a modem. A null-modem connection crosses transmit/receive and appropriate handshake lines to connect compatible DTEs directly.

226. What is GPIB/IEEE 488?

An instrumentation bus with 8 parallel data lines plus handshake/management lines. Devices act as talkers, listeners and controllers; the classic arrangement allows up to 15 connected devices including the controller.

227. What are noise, skew, setup time and hold time?

Noise is unwanted disturbance; skew is unequal signal arrival time. Setup/hold specify how long data must be stable before/after its sampling event.

13 / Keyboard, encoders, stepper motor and relay

Basic to expert

228. How does a matrix keyboard work?

The controller selects rows and reads columns to identify pressed intersections, reducing required connections compared with one wire per key.

229. What does a keyboard encoder do?

It detects a pressed key and produces its assigned code. Scanning and debouncing may be performed by dedicated hardware or software.

230. What is switch bounce?

Mechanical contacts rapidly open/close briefly during a transition. Debouncing prevents one physical press from being registered several times.

231. What is keyboard ghosting?

Certain combinations of pressed keys create an apparent extra key through shared matrix paths. Per-key diodes or suitable rollover restrictions can prevent it.

232. What is a stepper motor?

A motor that moves through discrete angular steps when its windings are energized in a controlled sequence.

233. How do you control its direction, speed and position?

Reverse the phase sequence for direction, change the step rate for speed and count steps for nominal position, assuming no missed steps.

234. How many steps per revolution for a 1.8-degree motor?

360/1.8=200 full steps. At 1000 steps/s, ideal speed is 1000/200 x 60=300 rpm.

235. Why not connect motor windings directly to a PPI pin?

Their current and inductive voltage transients exceed normal logic-pin capabilities. Use an appropriate driver and protection circuitry.

236. Why can an open-loop stepper lose position?

Excess load, too high a step rate or abrupt acceleration can cause missed steps; counting commands does not prove actual movement.

237. What is an optical shaft encoder?

A light source and detector sense a rotating patterned disc to produce position or movement signals.

238. What is absolute versus incremental encoding?

An absolute encoder gives a code identifying position within its specified range. An incremental encoder gives motion pulses and usually needs a reference to establish absolute position.

239. Why is Gray code used in some absolute encoders?

Adjacent positions change only one bit, reducing ambiguous readings during transitions between positions.

240. How is direction detected with quadrature signals?

Channels A and B are phase-shifted; which channel leads indicates rotation direction. Counting selected edges determines resolution.

241. What is a relay?

An electrically operated switch. In an electromechanical relay, a coil creates magnetic force that changes the contact state.

242. Why use a flyback diode across a DC relay coil?

It provides a path for inductive current when the driver turns off, limiting the voltage spike. Its polarity is normally reverse-biased while the coil is powered.

14 / Data entry and display devices

Basic to intermediate

243. What are OMR, OCR and MICR?

OMR detects marks in predefined positions; OCR recognizes printed or handwritten characters; MICR reads characters printed with magnetic ink, commonly on cheques.

244. How does a barcode reader work?

It senses light reflected from bars and spaces, then decodes their pattern according to a barcode symbology. A database often supplies the associated product information.

245. Is a barcode normally the product price itself?

Usually it is an identifier used to retrieve price/details. Some barcode formats can encode price or weight, so the answer depends on the application.

246. How does an image scanner work?

It illuminates a document, samples reflected light with sensors and converts the signals into digital image data.

247. What is a digitizer?

An input device that converts positions, drawings or physical measurements into digital coordinates/data, such as a graphics tablet.

248. What is a CRT display?

A cathode-ray tube directs an electron beam onto a phosphor screen. Beam scanning and intensity modulation form the image.

249. What is raster versus vector display?

Raster displays represent an image as a grid of pixels scanned or addressed across the screen; vector displays draw selected lines directly.

250. How does an LCD work?

Liquid-crystal cells alter light transmission through polarizers. Conventional LCDs modulate a backlight rather than emitting their own light.

251. Is every LED monitor an emissive LED-pixel display?

No. The term often means an LED-backlit LCD. Direct-view LED displays and OLED displays use different emitting-pixel technologies.

252. What are resolution, refresh rate and colour depth?

Resolution is the pixel grid; refresh rate is image updates per second; colour depth is the number of bits used to represent pixel colour values.

253. How much memory for 640x480 at 8 bits per pixel?

640 x 480 x 8/8 = 307,200 bytes, or 300 KiB, excluding padding and additional buffers.

15 / Printers, plotters and storage

Basic to intermediate

254. What is impact versus non-impact printing?

Impact printers strike an inked ribbon against paper; non-impact printers use other processes such as ink droplets, toner or heat.

255. How does a dot-matrix printer work?

Pins strike a ribbon to form dots on paper. It can produce multipart forms but is noisy and has limited image quality compared with many non-impact printers.

256. How does a daisy-wheel printer differ?

It strikes preformed characters on a wheel, giving letter-style output but limited graphics and a fixed character set per wheel.

257. How does an inkjet printer work?

It ejects small ink droplets onto paper, commonly using thermal or piezoelectric actuation.

258. How does a laser printer work?

A charged photoconductive surface is exposed to form a latent image, toner develops the image, toner transfers to paper and heat/pressure fuse it; the surface is then cleaned/reconditioned.

259. What is toner versus ink?

Toner is fine printing powder used in electrophotographic printers; ink is liquid colourant used in inkjet and other printing processes.

260. What does CMYK stand for?

Cyan, Magenta, Yellow and Key/Black. Cyan is not green, and magenta is not simply red.

261. What is DPI?

Dots per inch, a printer/scanner spatial-resolution measure. It is different from pages per minute, which measures printing speed.

262. What is a plotter and what are its two classic types?

A plotter produces drawings, traditionally with a pen. A flat-bed plotter moves the pen over stationary paper; a drum plotter moves paper in one axis and the pen in the other.

263. How does a magnetic hard disk store data?

Magnetic patterns on rotating platters are written/read by heads. Encoding and signal processing recover data; it is more complex than assigning one pole directly to each binary digit.

264. What are track, sector and cylinder?

A track is a circular recording path; a sector is an addressed subdivision; a cylinder is the set of tracks at the same radius across platter surfaces in the classic model.

265. What determines hard-disk access time?

Seek time, rotational latency, transfer time and controller/queueing overhead. The exact contribution depends on the request and drive.

266. What is average rotational latency at 7200 rpm?

One revolution takes 60/7200 seconds=8.33 ms. The ideal average half-turn wait is approximately 4.17 ms.

267. What is a floppy disk?

A flexible magnetic storage medium inside a protective enclosure, read/written by a floppy drive.

268. How is a CD read?

A laser/optical pickup detects changes associated with the recorded track, and decoding reconstructs data. Avoid equating an entire pit directly with a user-data 0 or 1.

269. How does SSD storage differ from HDD storage?

An SSD stores data in nonvolatile semiconductor memory and has no moving heads/platters; an HDD stores data magnetically on spinning media.

270. What are RAM, ROM and secondary storage?

RAM is working read/write memory, usually volatile; ROM/nonvolatile firmware memory holds persistent code/data; secondary storage retains larger files/programs across power loss.

16 / Fast oral challenges

Expert

271. If AL=FFH, does NOT AL set ZF because the result is zero?

No. AL becomes 00H but NOT does not change ZF or any other flag.

272. If AL=80H, does SF=1 prove that an error occurred?

No. SF reports the top bit. Whether the value is valid depends on signed/unsigned interpretation and the operation; check OF or CF for the relevant condition.

273. Can OR be used to add two numbers?

Not in general. It coincides with addition when there are no overlapping 1 bits that produce carries, but it is fundamentally a logical operation.

274. How could you swap two register values?

XCHG AL,BL is clear and direct. XOR-swap sequences exist but are not necessary and introduce pitfalls when the operands are not independent.

275. How do you test whether an unsigned byte is odd?

TEST AL,01H. ZF=0 means bit 0 is set, so the number is odd; ZF=1 means even.

276. Can two different segment:offset pairs refer to the same address?

Yes. 1000H:0010H and 1001H:0000H both form physical address 10010H.

277. Can an 8086 word start at an odd memory address?

Yes, but accessing a word at an odd address requires two bus transfers on the 8086 rather than the usual aligned one-transfer case.

278. Why do 8086 systems use ALE and latches?

The lower address and data signals share multiplexed pins. ALE lets an external latch retain the address while those pins later carry data.

279. What is the 8086 versus 8088 bus difference?

Both have a 16-bit programming architecture, but the 8086 has a 16-bit external data bus and a six-byte queue; the 8088 has an 8-bit external data bus and a four-byte queue.

280. What is a safe answer if sir asks an unverified board-specific detail?

State the principle, identify the value you can see in the approved program or board label, and say which schematic/jumper setting determines the remaining detail. Do not invent a port address or DAC voltage.

17 / Additional questions from the class notes

Intermediate to expert

281. What is N-key rollover?

The keyboard can correctly register multiple simultaneous keys, up to its supported rollover capability. It differs from lockout, which suppresses ambiguous additional keypresses.

282. What is key lockout?

A keyboard policy that inhibits additional key detection under specified multiple-key conditions, avoiding ambiguous codes at the expense of simultaneous-key entry.

283. What is an electroluminescent display?

A display whose emitting material produces light under electrical excitation. Unlike an ordinary LCD, the emitting layer itself generates light.

284. What is the difference between a character display and a graphic display?

A character display presents defined character cells/glyphs; a graphic display allows individual pixels or graphical primitives to form arbitrary images.

285. What is persistence in a CRT?

The time phosphor continues emitting after excitation. It affects flicker and motion smearing and is different from refresh rate.

286. How does a thermal printer work?

A heated printhead creates marks on heat-sensitive paper or transfers material from a ribbon, depending on the thermal-printing method.

287. How does a drum printer work?

A rotating drum carries character shapes; timed hammers press paper and ribbon against the selected characters. It is an impact line-printing mechanism.

288. How does a chain printer work?

Character shapes move on a chain or band, and timed hammers strike as required characters pass each print position.

289. What are character, line and page printers?

These categories describe the unit printed or processed at a time: one character, a line, or a page image. They are distinct from impact/non-impact classification.

290. What is spooling?

Storing print jobs in a queue so applications can continue while the printer processes jobs at its own speed.

291. What is WORM storage?

Write Once, Read Many: data can be recorded once and read repeatedly; existing recorded content cannot be normally overwritten.

292. How do CD-ROM, CD-R and CD-RW differ?

CD-ROM is prerecorded read-only media; CD-R is recordable once at a location; CD-RW supports erasing and rewriting using a suitable recording mechanism.

293. What is the role of an 8251A transmitter?

It accepts parallel data from the CPU and serializes it with the selected synchronous or asynchronous framing.

294. What is the role of an 8251A receiver?

It accepts serial bits, assembles characters, checks applicable errors and presents received data to the processor.

295. What are the major 8251A functional blocks?

Data bus buffer, read/write control logic, transmitter, receiver and modem-control functions.

296. How do a mode instruction and command instruction differ in a USART?

A mode instruction establishes the communication format; a command controls operations such as transmitter/receiver enable and error reset. Follow the device's initialization sequence.

297. What is a shift register's role in serial communication?

It converts parallel data to serial bits for transmission and serial bits to parallel data for reception.

298. How would you interface an ADC through an 8255?

Configure the data port as input; use suitable control outputs to start conversion and status inputs to detect completion; then read the valid result. Pin polarity/timing comes from the ADC datasheet.

299. What are sampling and quantization?

Sampling selects signal values at discrete times; quantization maps amplitudes to a finite set of digital levels. They are separate steps in analog-to-digital conversion.

300. What is the Nyquist condition and why use an anti-alias filter?

For an ideally band-limited signal, sample faster than twice its highest frequency. A practical anti-alias filter attenuates unwanted frequencies before sampling; real systems allow transition-band margin.

Appendix A | Practical result sheet

Exp.	Operation	Bytes to enter	Result
1	01H + F0H	B0 01 04 F0	AL=F1H
2	07H - 02H	B0 07 2C 02	AL=05H
3	02H AND A3H	B0 02 24 A3	AL=02H
4	02H OR A3H	B0 02 0C A3	AL=A3H
5	NOT 02H (8-bit)	B0 02 F6 D0	AL=FDH
6	02H XOR A3H	B0 02 34 A3	AL=A1H
7	02H SHL 1	B0 02 D0 E0	AL=04H
8a	02H SHR 1 (contents)	B0 02 D0 E8	AL=01H
8b	05H SHR 1 (worked)	B0 05 D0 E8	AL=02H

These are the instruction bytes, not a complete universal run-and-return program. Use the trainer monitor's approved execution/stop method.

Result/operation	CF	OF	SF	ZF	PF	AF
01+F0=F1	0	0	1	0	0	0
07-02=05	0	0	0	0	1	0
02 AND A3=02	0	0	0	0	0	U
02 OR A3=A3	0	0	1	0	1	U
02 XOR A3=A1	0	0	1	0	0	U
02 SHL 1=04	0	0	0	0	0	U
02 SHR 1=01	0	0	0	0	0	U
05 SHR 1=02	1	0	0	0	0	U
NOT 02=FD	P	P	P	P	P	P

U = undefined; P = preserved, not recomputed. PF uses even parity of the low result byte. All shift rows use count one.

Appendix B | Hardware quick reference

Register	Address in the program tables	Typical write
8255 control	1FH	80H: all ports output, Mode 0
Port A	19H	Segment byte or pattern
Port B	1BH	Selection/colour/pattern as wired
Port C	1DH	Selection/control/pattern as wired

Follow the table-based program only on a matching trainer map. 19H-1FH and the 3FDx sketch are not interchangeable.

Experiment	Writes after control=80H	What changes
9: seven segment	B=F0H, C=00H, A=segment byte	A selects the glyph under the given mapping
10: LED display	A=FFH, C=00H, B=F0H	Modify the indicated pattern and use the demonstrated jumper setup
11: D/A title	A=FFH, B=F0H, C=00H	The note varies C; analog meaning requires circuit confirmation
12: LED matrix	A=FFH, B=00H, C=00H	Later sketch uses C=01H; row/colour mapping depends on wiring

Standard seven-segment pattern table

Assumption: active-low segments; bit 7=dp, bit 6=g ... bit 0=a. Decimal point off.

Glyph	0	1	2	3	4	5	6	7
Hex	C0	F9	A4	B0	99	92	82	F8
Glyph	8	9	A	b	C	d	E	F
Hex	80	90	88	83	C6	A1	86	8E

Example derivation: digit 0 lights a,b,c,d,e,f. With active-low wiring, bits 0-5 are 0, g and dp are 1: 11000000B=C0H.

Essential formulas

Physical address = segment x 16 + offset.

Eight-bit two's complement of x = (256 - x) mod 256.

Ideal DAC step = Vref / 2^n; ideal Vout = D x Vref / 2^n, under the stated voltage-scale convention.

Stepper steps/revolution = 360 / step angle; rpm = steps/second x 60 / steps/revolution.

8N1 character rate = serial bit rate / 10; eight-row frame rate = row-update rate / 8.

Appendix C | Rehearse these aloud

Explain one arithmetic experiment without reading the code.

Show the binary calculation for AND, OR and XOR of 02H and A3H.

Explain why CF and OF differ using FFH+01H and 7FH+01H.

Decode B0 02 D0 E8 and predict AL and CF.

Explain every bit of 80H as an 8255 control word.

Distinguish a control word, a port address and a segment pattern.

Derive C0H for zero from the stated wiring.

Explain why a static output does not automatically count 0-F.

Describe DAC conversion without claiming an unsupported voltage.

Explain LED-matrix scanning and how you would remove ghosting.

Diagnose a blank display in a logical order.

Explain polling, interrupts and DMA using a slow peripheral.

A model 30-second answer: addition

The aim is to add two 8-bit hexadecimal values on the MTS86 trainer. MOV AL,01H loads the first operand; ADD AL,0F0H adds the second. The bytes are B0 01 04 F0. The result is F1H in AL. There is no unsigned carry and no signed overflow. I verify the entered bytes, single-step the two instructions and inspect AL and the flags.

A model 30-second answer: seven segment

The aim is to display a hexadecimal character through the interface. In the program table, I write 80H to control port 1FH to configure Mode 0 outputs, then set the selection ports and write a segment pattern to Port A at 19H. With the stated active-low segment mapping, C0H lights a-f to show zero. I confirm the trainer map and jumpers because the scan includes a conflicting sketch.

When you do not know an exact detail

Explain the principle first and say what determines the missing value: for example, "The DAC output depends on the input code and reference. I would check the actual reference and output circuit before stating a voltage." This is stronger than guessing.

Technical checks and references

The uploaded materials define the scope. Standard technical clarifications were cross-checked against manufacturer-authored documentation. The arithmetic/flag and segment-pattern examples were independently calculated. These references do not resolve the conflicting board address sketch.

Intel, The 8086 Family User's Manual (October 1979), university-hosted manufacturer manual

Open reference

Renesas/Intersil, 82C55A datasheet (8255-compatible CMOS interface)

Open reference

Texas Instruments, DAC0808 datasheet

Open reference

Source boundaries

The Intel reference was used to check instruction semantics and encoding. The Renesas reference was used to check compatible PPI modes/control format; device-specific enhancements are not assumed for an original 8255. The TI reference supports the conditional DAC0808 current-output example, not identification of the trainer's actual DAC.

Final self-check

Can you explain the aim, instructions, bytes, result, flags and verification for all eight software experiments? Can you explain configuration, data pattern, physical output and likely failure causes for all four hardware experiments? If yes, you have covered the core of this practical viva.